Chris Wu

(613) 879-2755 | chris.wu1@uwaterloo.ca | linkedin.com/in/chriswu2222 | github.com/ChrisWu0000 | chriswu.tech

Highlights

- **Academic Excellence:** Ranked Top 3 in University of Waterloo Computer Engineering cohort (4.0 GPA); recipient of the President's Scholarship of Distinction
- **Systems & AI Automation:** Built autonomous Python-based triage agents, LLM tool-calling workflows, and distributed CI/CD integration pipelines at Ciena and Nokia
- **Embedded Systems & Infrastructure:** Experience validating high-speed optical transceivers (WL6n), CMIS telemetry, and DPI/DDoS mitigation platforms across Linux environments
- **Industry Certifications:** CompTIA Security+ (SY0-701), Cisco CCNA, and Microsoft Azure Fundamentals (AZ-900) covering network infrastructure, threat response, and cloud architecture
- **Security Clearance:** Active Government of Canada Reliability Status clearance; trained in protocols for handling protected information

Education

University of Waterloo Sep. 2024 - Present
Bachelor of Applied Science in Computer Engineering (Co-op) *Waterloo, ON*

- **First in Class Engineering scholarship**, President's Scholarship of Distinction, 4.0 GPA
- **Relevant Coursework:** Algorithms and Data Structures, Probability Theory and Statistics, Discrete Mathematics and Logic, Chip Design and Tape Out, Embedded Microprocessor Systems, Systems Programming and Concurrency, Digital Computers, Electronic Circuits

Technical Skills

Languages: Python, C, C++, Bash, TCL, VHDL, Java, JavaScript, TypeScript, HTML, CSS
Frameworks/Libraries: NumPy, Pandas, LiteLLM, PyTorch, TensorFlow, Pytest, Pygame
Systems/Tools: STM32, Ubuntu, Docker, Wireshark, Git, GitHub, GASH, PyShell
Cloud/DevOps: Jenkins, Microsoft Azure, Cisco CLI, Nokia MDCLI, VS Code, PyCharm

Experience

Software Developer May 2026 – Aug. 2026
Ciena – Embedded Software Integration Testing Team *Ottawa, ON*

- Automated test cases for WaveLogic 6 Nano (WL6n) 400G–800G coherent optical transceivers to validate CMIS compliance, optical telemetry, and firmware states across 10+ **Jenkins CI/CD** pipelines
- Developed agentic **A/I workflows** in Python using **LiteLLM** to diagnose embedded test failures, query build logs across Jenkins, and synthesize root-cause summaries across internal developer tools
- Optimized **embedded regression** testing across OSFP800/QSFP-DD800 hardware testbeds, reducing CI suite runtime by **25%** through parallelized execution and **dynamic Jenkins pipeline scheduling**
- Engineered an automated triage pipeline using **Jira and Jenkins REST APIs** to extract register dumps from failed builds, auto-generating structured bug reports to **cut triage latency by 30%**
- Created real-time OpenSearch observability dashboards and automated Confluence sync pipelines to track **30+ physical-layer and firmware health metrics** across release builds

Quality Assurance Engineer Sep. - Dec. 2025
Nokia - Application Assurance and Defender Mitigation System Teams *Ottawa, ON*

- Worked on **leading-edge application identification** technology for risk mitigation of DDoS attacks
- Verified detection mechanisms to identify popular applications on Nokia 7750 SR series of IP routers

- Developed automated test cases to validate new application signatures; **modified and updated 80+ test automation scripts** programmed in TCL, Bash, and Python
- Leveraged **Pytest and PyShell** in a Linux-based regression environment, translating and upgrading legacy test scripts to Python-based platforms to improve testing readability, and maintenance
- Routinely performed software and scripts source code and version control using Git-style toolkits

Azure Application Developer Jan. - Apr. 2025
MD Pharma *Waterloo, ON*

- Engineered a prototype AI wellness application leveraging Microsoft Azure Cognitive Services to process and analyze user health inputs
- Integrated Azure Text Analytics and Language Understanding (LUIS) models to extract key medical entities and user intents, benchmarking performance against 35 datasets to achieve a 91% accuracy rate
- Applied sentiment analysis to unstructured natural language data to accurately evaluate user emotional context and health input severity, enabling dynamic, context-aware responses tailored each patient
- Conducted comprehensive market research on emerging AI health technologies and competitor platforms, synthesizing findings into structured technical reports and presentations

Projects

Waterloo Rocketry | *Python, Java, NumPy, ML* Sep. 2024 - Aug. 2026

- Directed the software subteam as Software Lead, coordinating the development of the flight simulation engine and telemetry pipelines through rigorous code reviews and cross-functional hardware integration
- Engineered a Python flight simulation engine to model aerodynamics, propulsion curves, and apogee targets, directly supporting the world-record amateur liquid rocket launch reaching 63,497 ft
- Architected and maintained telemetry processing pipelines in Python and Java to ingest, decode, and store real-time sensor streams from critical rocket subsystems
- Delivered core analytics features to the Omnibus ground station dashboard to provide live trajectory tracking and visualization during launch operations

WatRooms | *C++, STM32, Bluetooth, Git*

- Designed and prototyped a campus study room access device using Bluetooth communication and student number authentication for the Project Studio course
- Developed a hardware booking and authentication system, integrating C++ on STM32 microcontrollers
- Utilized HC-05 Bluetooth modules to manage secure communication over a Personal Area Network

Miedema's Musical Massacre | *Python, Pygame, Git*

- Led a team of 3 to design and program a desktop game using Pygame within a strict 3-month timeline
- Managed all phases of the software development life cycle (SDLC): requirements gathering, high-level architecture design, implementation, troubleshooting, and engineering technical documentation

Certifications

- **CompTIA Security+ (SY0-701)** – Validated expertise across enterprise security architecture, vulnerability management, threat response, and incident remediation operations (Obtained Sep. 2026)
- **Cisco Certified Network Associate (CCNA)** – Validated network access, IP connectivity, IP services, and security fundamentals. Completed configuration labs for Cisco routers/switches including VLANs, trunking, ACLs, and OSPF (Obtained Apr. 2025)
- **Microsoft Azure Fundamentals (AZ-900) & Azure AI Fundamentals (AI-900)** – Demonstrated foundational knowledge of cloud concepts, core Azure governance tools, and cloud-driven AI solution development (Obtained Feb. 2025)